ENTERPRISE DATA MANAGEMENT (EDM)

INDIVIDUAL INTEGRATED CAPSTONE ASSIGNMENT: 30% OF FINAL COURSE GRADE

1. ASSIGNMENT OVERVIEW

Item	Details
Assignment Title	*Project Savanna: Enterprise Data Management Turnaround*
Type	Individual portfolio-based capstone (NO team components)
Weight	30% of final grade
Duration	1 week, with 2 formative checkpoints
Total Marks	100
Submission	Written portfolio (PDF via Moodle/Turnitin) + GitHub repository of technical artifacts + published dashboard link + 15-minute individual viva defense
Competencies Assessed	C1, C2, C3, C4, C5, C6, C7, C8, C9 (ALL)

Your Role: You have been hired as the first-ever Enterprise Data Management Consultant for Savanna Retail Group (SRG), a mid-sized Ugandan omnichannel retailer in serious data trouble. The Board has given you 18 months and a fixed budget to transform the organization from "data chaos" to "data-driven." Your entire portfolio is the professional consultancy package you would deliver to the Board.

This is deliberately an ill-structured problem. There is no single correct answer. You will be graded on the quality of your reasoning, the authenticity of your hands-on artifacts, and the defensibility of your decisions, and not on matching a "model answer."

2. COMPETENCY COVERAGE MAP

Part	Task	Competency
A	A1 – EDM Diagnostic & Data Lifecycle Mapping	C1

A	A2 – Governance Framework & Compliance Register	C2
B	B1 – Architecture Selection, ER Modeling & Normalization	C3
B	B2 – Data Quality Assessment & Cleansing (hands-on)	C4
B	B3 – Master Data Management Design	C5
B	B4 – ETL/Integration Design + Real-Time Architecture (hands-on)	C6
C	C1 – Metadata Dictionary, Lineage & Catalog Strategy	C7
C	C2 – Security Risk Assessment, Access Controls & Privacy Compliance	C8
C	C3 – BI Dashboard Build + Emerging Technology Evaluation	C9
D	D1 – Integrated 18-Month EDM Strategy Roadmap	Synthesis (C1–C9)
D	D2 – Decision Log, Red-Team Critique & Reflective Essay + Viva	Deep thinking (C1–C9)

3. THE SCENARIO: SAVANNA RETAIL GROUP (SRG)

3.1 Organizational Profile

- Ugandan-headquartered omnichannel retailer: 23 physical stores (Kampala, Jinja, Mbarara, Gulu), an e-commerce platform ("SavannaShop"), and a loyalty program with 180,000 registered members.
- Integrates MTN Mobile Money and Airtel Money for payments.
- Expanding into Kenya and Rwanda, and negotiating a supply partnership with a European distributor whose due-diligence team requires GDPR-equivalent data practices. The Uganda Data Protection and Privacy Act (DPPA), 2019 and PDPO registration obligations apply now.
- Recent incident: A laptop containing an unencrypted export of 9,000 customer records (names, phone numbers, purchase history) was stolen from the Jinja store. It was reported internally 11 days late. No breach notification was made.

3.2 Current Systems Landscape (all siloed)

1. Legacy POS: separate MySQL database per store (23 databases)

2. E-commerce: PostgreSQL
3. Loyalty program: standalone mobile app with its own cloud database
4. CRM: SaaS platform
5. ERP/Inventory: Odoo at HQ only
6. HR & Payroll: local system + Excel spreadsheets
7. Supplier records: 14 different Excel files from procurement officers
8. Mobile money reconciliation: manual CSV downloads

3.3 Reported Problems

- Estimated 23% duplicate customer records across systems; loyalty points issued to wrong members
- Product codes inconsistent between POS, e-commerce, and supplier files (same product, 5 identifiers)
- Monthly management reporting takes 11 days of manual Excel consolidation
- Stock discrepancies: book stock vs. physical stock variance of 8.4% — nobody knows why
- Stock-outs in high-demand stores while slow stores hold excess inventory
- No one can answer the Board's question: "Which customer segments are growing, which are churning, and what should we do about it?"

3.4 Your Constraints (these are REAL and non-negotiable — your recommendations must respect them)

- ✓ Annual data budget: UGX 480,000,000 (~USD 130,000) including licenses, cloud costs, and any new hires
- ✓ IT team: 4 people (only 2 can write SQL; zero data engineering experience)
- ✓ Intermittent store connectivity — several stores lose connectivity for hours daily
- ✓ Board patience: 18 months — they expect visible quick wins by month 6
- ✓ PDPO audit expected within 12 months
- ✓ Power reliability issues at some stores (UPS-backed POS only)

3.5 Instructor-Provided Resources

File	Contents
SRG_Customers.csv	5,000 records — duplicates, missing values, mixed phone/date/gender formats, invalid districts
SRG_Products.csv	1,200 records — duplicate SKUs, unit inconsistencies (kg/KGS/Kilograms), category typos
SRG_Sales.csv	60,000 transactions across 6 months in 3 inconsistent formats (POS / e-commerce / app exports), negative quantities (returns), mixed currency notations, null payment refs
SRG_Suppliers.csv, **SRG_Loyalty.csv**, **SRG_HR.xlsx**	Additional sources; HR file contains sensitive fields you must identify
etl_error_log.txt	Log from a "broken" nightly ETL job — you will diagnose it (Task B4)
footfall_stream_sample.json	Simulated in-store sensor events for the real-time design task
Templates	Assumptions Register, Decision Log, RACI matrix, Risk Matrix, DPIA template (Appendices A–E of this brief)

4. REQUIRED DELIVERABLES

PART A — FOUNDATIONS & GOVERNANCE (15 marks)

Task A1 — The EDM Diagnostic (C1) — 5 marks

Produce a 2-page Executive Briefing Memo to the Board plus one Data Lifecycle Map diagram covering:

1. A precise differentiation of enterprise data management vs. traditional database management, using SRG evidence (the Board currently believes "hiring two more DBAs will fix this" — dismantle that argument with evidence).

2. A data lifecycle diagram for customer data — from creation (POS signup, loyalty registration, e-commerce checkout, mobile money linkage) through storage, usage, sharing, archiving, and deletion — identifying the system, the stakeholder, and the failure point at every stage.

3. A mapping of SRG's problems onto the DAMA-DMBOK knowledge areas (wheel), showing which functions SRG lacks entirely.

Think deeper: Why do the same symptoms (duplicates, stock variance, slow reporting) keep returning after every "clean-up exercise"? What does that tell you about the difference between treating data as a by-product versus a managed asset?

Task A2: Governance Framework & Compliance Register (C2), 10 marks

Deliver:

i. A Data Governance Charter (vision, principles, scope) appropriate to SRG's size and maturity do not copy an enterprise framework designed for a bank.
ii. An operating structure: council, stewardship, and ownership roles, populated with SRG job titles (existing staff only you cannot hire 6 people).
iii. A RACI matrix covering at least 10 concrete data activities (e.g., customer record creation, master data change approval, breach declaration, retention execution).
iv. Three enforceable policies written in implementable language: (a) Data Access & Classification, (b) Data Quality, (c) Data Retention & Deletion. Each policy must have named owners, measurable requirements, and enforcement mechanisms.
v. A Compliance Obligations & Gap Register: map SRG's current practices against the Uganda DPPA 2019 (cite specific sections), GDPR (cite specific articles relevant to the EU partnership lawful basis, data subject rights, breach notification, cross-border transfer), and CCPA as a comparative reference. Prioritize gaps by risk.
vi. A 1-page memo rebutting a skeptical CFO who calls governance "bureaucracy" quantify the cost of inaction using the stolen-laptop incident and at least two comparable industry cases.

Think deeper: Which single governance policy, if enforced from day one, would have prevented the Jinja laptop incident? Which would be violated daily by SRG's own staff habits and how would you make it enforceable rather than decorative?

PART B — TECHNICAL CORE (40 marks)

Task B1: Architecture & Data Modeling (C3). 8 marks

i. Architecture pattern decision: Evaluate centralized, decentralized, federated, and hub-and-spoke patterns against SRG's constraints (budget, connectivity, skills, 23 stores). Recommend one with a weighted trade-off table — include at least 5 evaluation criteria and justify the weights.

ii. Construct a conceptual and logical ER model (draw.io/Lucidchart) for the consolidated operational core: Customer, Product, Store, Order, OrderLine, Payment, Supplier, LoyaltyAccount. Use Crow's Foot notation; show cardinality and key attributes.

iii. Take the provided unnormalized sales extract and document normalization through 1NF → 2NF → 3NF, showing each step with rationale.

iv. Propose one justified denormalization for reporting purposes, with an explicit performance-vs-integrity trade-off memo.

Think deeper: Your ER model embeds assumptions (e.g., "a customer may belong to a household"). List three such assumptions and describe a real SRG situation where each assumption breaks.

Task B2: Data Quality Assessment & Cleansing (C4) 10 marks: HANDS-ON

This is an execution task. Using SQL, Python, or Excel, on the provided dirty datasets:

1. Profile `SRG_Customers.csv` and `SRG_Products.csv` and compute quantitative metrics for all five dimensions accuracy, completeness, consistency, timeliness, validity plus uniqueness. Examples of required outputs: % duplicate persons (define your matching logic), % missing contact fields, phone format variant counts, invalid district entries, date-format variant counts.
2. Execute a cleansing plan (scripts must be in your GitHub repo): standardization, deduplication, enrichment, correction and produce a before/after metrics table proving improvement (e.g., duplicates 23.1% → 2.4%).
3. Write a catalog of at least 12 validation rules (technical + business rules) that would prevent recurrence at the point of entry.
4. Perform a root cause analysis of the 8.4% stock variance using a fishbone diagram distinguish systemic causes from point-of-entry causes.

5. Design a continuous data quality monitoring specification: metrics, thresholds, alert triggers, escalation paths, owners.

Think deeper: Do not just report "18% of phones are malformed." Explain why which upstream process, system, or human behavior produces each defect class and what process change (not just data fix) stops it. Causal reasoning earns marks; description does not.

Task B3: Master Data Management Design (C5) 10 marks

Design SRG's Customer MDM solution:

1. Select and justify the MDM architectural style (registry, consolidation, coexistence, centralized) against SRG's constraints include a rejected-alternatives analysis.
2. Design the golden record schema with attribute-level survivorship rules (which system wins for name, phone, address, email, preferences and why, with business justification).
3. Specify matching rules: deterministic keys plus a probabilistic weighted-scoring approach (e.g., name similarity, phone, district); state your score thresholds for auto-merge vs. manual review and justify the trade-off between false merges and missed matches. Address the ethical risk of incorrect merges (e.g., merging two customers destroys one person's loyalty history and privacy rights).
4. Define master data hierarchies (customer→household where defensible; product→category→subcategory) and show how the golden record integrates back into POS, e-commerce, loyalty, and CRM business processes (integration flows diagram).
5. Define the MDM governance process: who approves new attributes, who resolves stewardship disputes, how changes propagate.

Think deeper: A store manager refuses to stop creating "quick" local customer records because network outages block the central system. Resolve this conflict technically and politically without breaking your MDM design or ignoring the connectivity constraint.

Task B4: Data Integration & Real-Time Architecture (C6) 12 marks — HANDS-ON

1. Design a star-schema data warehouse (dimensions/facts diagram) fed by all five transactional sources.

2. Document a complete ETL workflow: extraction method per source (given connectivity reality), transformation rules (data type conversions, deduplication, lookup enrichment, Slowly Changing Dimension Type 2 for product/store history), and load sequencing, delivered as a visual workflow diagram plus implemented core transformations in SQL (or Talend, or fully documented pseudo-code) in your repo.

3. Produce an ETL vs. ELT decision matrix (latency, cost, scalability, complexity, SRG fit) for a future cloud migration.

4. Diagnose the provided broken ETL job (`etl_error_log.txt`): identify the root cause(s), document your reasoning trail, and prescribe corrective actions plus prevention.

5. Design a real-time integration architecture using the footfall sensor stream: specify streaming components (e.g., CDC + message broker + consumer), processing logic, and output destinations targeting two use cases: live stock visibility and mobile money fraud pattern detection. Justify batch vs. streaming per use case rather than defaulting to "real-time for everything."

Think deeper: The Board wants "real-time everything." Argue where real-time genuinely creates business value at SRG and where it would burn the entire budget for no return. Precision here demonstrates executive-level judgment.

PART C — GOVERNANCE TECHNOLOGY, PROTECTION & INSIGHT (25 marks)

Task C1: Metadata Management (C7) 7 marks

1. Build a data dictionary (business + technical + administrative metadata) for at least 25 warehouse elements.
2. Produce a lineage diagram tracing one board-level KPI (e.g., "monthly active customers") from raw source fields through every transformation to the dashboard — annotated at each hop.
3. Design a tagging taxonomy for SRG's data catalog (domain tags + sensitivity classification tags + quality tags) and apply it to your dictionary.

4. Perform an impact analysis: a proposed change to the customer address schema — which reports, dashboards, and processes break? Document dependencies.
5. Compare two metadata/catalog tools (e.g., Apache Atlas vs. a commercial option) against SRG-defined criteria and recommend one with budget justification.
6. Define the metadata governance process (who updates, review cycles, integration with Task A2 governance).

Task C2 Security & Privacy (C8) 10 marks

1. Threat model SRG's data estate (insider threats, external attack, accidental exposure — including the store-connectivity and stolen-device realities).
2. Conduct a risk assessment: minimum 10 risks scored on a likelihood × impact matrix, with mitigations.
3. Design an RBAC access control matrix (roles × data objects × operations) — and implement it: submit working SQL `GRANT`/`REVOKE` scripts with test evidence showing an attempted unauthorized access fails.
4. Specify encryption and masking requirements (at-rest, in-transit, key management) and demonstrate data masking of sensitive fields in your analytics copy of the dataset (show the technique).
5. Complete a DPIA (using the template) for the loyalty app's EU-facing rollout.
6. Produce a breach response plan — rewrite SRG's failed response to the Jinja laptop incident into a compliant procedure under both DPPA 2019 and GDPR timelines.
7. Deliver a compliance audit checklist result: SRG's current state against your Task A2 obligations register, with a prioritized remediation plan.

Think deeper: Full encryption of everything is the "safe" answer but not a strategy. Allocate SRG's limited security budget across the risk register — defend every omission as explicitly as every inclusion.

Task C3: Analytics, BI & Emerging Technology (C9) 8 marks — HANDS-ON

1. Formulate 5 strategic business questions the Board actually needs answered (churn, segment growth, stock optimization, fraud, expansion).

2. Design the analytics workflow for these questions: data sources → preparation → analytical methods → outputs, with rationale per decision.
3. Build and publish an interactive dashboard (Power BI or Tableau Public) from the cleansed/integrated dataset — must include filters, multiple visualization types, and drill-down. Submit the published link + screenshots.
4. Write a 1-page Executive Insight Brief: 3 findings from your dashboard and the recommended business actions.
5. Evaluate emerging technology for SRG: one cloud EDM platform stack (e.g., Azure/AWS/GCP data services or Snowflake/Databricks) and one AI-driven trend (AI-powered data quality, data mesh, or DataOps) — assessed against SRG's budget, skills, and connectivity constraints, with a clear adopt/pilot/reject verdict for each.
6. Assess SRG's analytics maturity using a provided framework and define the realistic 2-year advancement path.

Think deeper: Critique your own dashboard using Tufte/Few principles: which chart did you delete, why was it misleading, and what replaced it? Dashboards that merely "look nice" score below dashboards that survive scrutiny.

PART D — SYNTHESIS, DEFENSE & REFLECTION (20 marks)

Task D1 — The Integrated EDM Strategy Roadmap (8 marks)

Produce the board-ready 18-month implementation roadmap integrating ALL prior parts:

- Phased plan with milestones, dependencies, and governance checkpoints (with quick wins visible by month 6 — the Board requirement)
- Resource and budget allocation within the UGX 480M envelope (itemized — and show your arithmetic)
- Risk mitigation and success metrics/KPIs per phase
- An explicit coherence narrative: one page showing how governance feeds quality, quality enables MDM, MDM enables integration, metadata sustains it, security wraps it, and analytics monetizes it.

Task D2: Decision Log, Red Team & Reflective Essay (7 marks) + Viva Defense (5 marks)

1. Decision Log (template provided, minimum 8 entries): for each major decision record the decision, alternatives considered, evaluation criteria, evidence used, the choice, and what new evidence would make you change your mind.
2. Red-Team Section (min. 500 words): construct the strongest possible attack on your own strategy (budget, change resistance, connectivity, staff skills, political capture of governance), then rebut it or concede honestly where it wins.
3. Reflective Essay (800–1,000 words): connect your technical choices to Uganda's digital transformation context, data ethics (consent in loyalty programs, footfall sensor surveillance, algorithmic bias in matching rules, AI fairness), and your own intended career path (CDMP certification mapping included).
4. Viva (15 minutes, Week 15): individual defense. Expect "Why?", "What if the budget is halved?", and "Find the flaw in your own ETL design" questions. The viva verifies authorship and depth of reasoning artifacts you cannot explain will not earn their marks.

6. SUBMISSION REQUIREMENTS

Component	Format
Written portfolio	Single PDF, max 40 pages excluding appendices, professional consultancy formatting with executive summary
Code & technical artifacts	GitHub repo (or institutional repository): SQL/Python cleansing scripts, ETL code, access-control scripts, masking script
Diagrams	Embedded in portfolio + source files (draw.io/Lucidchart exports)
Dashboard	Live published link (Power BI / Tableau Public) + screenshots
Logs/evidence	Screenshots of profiling outputs, before/after metric tables, failed-access test evidence
Templates	Completed Assumptions Register, Decision Log, Risk Matrix, DPIA, RACI (appendices)

Integrity	Turnitin submission; declare any AI-tool assistance per KIU policy — undeclared use or inability to explain artifacts in the viva constitutes an integrity violation